DUAL MOTHERBOARDS

The Dual Motherboard configuration in the Monkey Head Project combines two high-performance platforms to maximize computational flexibility, redundancy, and specialization. Workloads are strategically assigned to leverage each board's strengths, ensuring optimal resource use and resilient operations.

Primary Motherboard — Supermicro X9QRI-F+:
- Quad Intel Xeon E5-4627 V2 CPUs for high-throughput, parallelized computation.
- Optimized for AI model training, large-scale simulation, and orchestration leadership tasks.
- ECC Registered memory for maximum stability in continuous uptime environments.

Secondary Motherboard — ASUS ROG Zenith Extreme Alpha:
- AMD Ryzen Threadripper 1950X with 16 cores / 32 threads for multi-threaded workloads.
- Tuned for real-time analytics, orchestration worker roles, and specialized GPU acceleration.
- High-bandwidth PCIe lanes for compute offload and expansion modules.

Shared & Supporting Resources:
- Hybrid ECC / non-ECC RAM pools tailored to workload type.
- Unified liquid cooling loop with thermal isolation per CPU zone.
- Dual independent PSUs with failover switching.

Governance Integration:
- All workload assignments, configuration changes, and performance benchmarks are logged in the Unified Audit Log (UAL).
- Resource reallocation requires governance approval if crossing defined operational thresholds.

Operational Advantages:
- Parallel specialization — AI model training can run alongside real-time operations without resource contention.
- High fault tolerance — Failure of one board does not compromise system operation.
- Scalable — Both boards can be expanded independently.

KPIs:
- Average CPU utilization split across boards.
- Job completion time reduction from parallelization.
- MTTR (Mean Time To Recovery) after simulated failure.

Testing Objectives:
- Benchmark inter-board communication latency and throughput.
- Validate automated workload reassignment during single-board outage.
- Stress-test thermal stability under maximum combined load.

Cross-References:
- See 'Hardware Architecture Overview' for complete system context.
- See 'Orchestration' for how workloads are assigned between boards.